# Identifiability by Stationarity: Blind Gain–Object Separation in Self-Calibrating Ghost Imaging

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***Abstract*—With known gain, an ordered orthogonal scan minimizes conditioning-driven amplification of additive noise; with unknown slow multiplicative gain and unconstrained objects, the same deterministic chronology creates an exact gain–object ambiguity. We resolve this paradox with an identifiability theory of blind gain–object separation for single-pixel (ghost) imaging, distinguishing algebraic identifiability, statistical identifiability, and estimation conditioning. Every square invertible design fails algebraically (modulo global scale); tall designs restore uniqueness generically at $N \geq K+p$ frames for a $p$-dimensional drift class. Independently, a stationarity anchor pins the relative gain statistically: once the object carrier is stationary in time, slow modulation of the record can only be gain, recovered at the minimax rate by a windowed log-domain estimator. Acquisition by the randomized Hadamard transform (SRHT, at full sampling) preserves orthogonal conditioning while supplying, under Walsh-flatness or a successful random permutation of a sufficiently spread object, the stationary or exchangeable carrier the estimator needs; an exact finite-noise relative mean-square-error identity then yields a design chart with a computable flip boundary between ordered and randomized acquisition, whose gain-known skeleton is validated on the simulation grid with no free parameters (raw-metric median agreement factor 1.04). In frame-matched simulations, raw ordered Hadamard leaves a non-decaying blind gain error of 0.94–0.99 across all averaging windows, where randomized arms reach 0.007–0.015 (medians with two-way clustered 95% confidence intervals), and randomized Hadamard acquisition recovers $+5.45$ dB under slow drift. The acquisition rule: randomize the carrier's chronology — make the object stop pretending to be time.**

***Index Terms*—ghost imaging; single-pixel imaging; blind calibration; identifiability; bilinear inverse problems**

## I. Introduction

Single-pixel (ghost) imaging [1], [2], [3], [4] has become the method of choice for imaging through media that defeat conventional cameras [5], [6]: turbid water, fog, dynamic scattering media [7], complex channels around corners [8]. The instrument is disarmingly simple — structured illumination patterns and a bucket detector [9], [10] — and the textbook design instinct is equally simple: use an ordered orthogonal basis. A Hadamard scan acquires one transform coefficient per frame [11], [12], the measurement matrix is perfectly conditioned, and an exact inverse waits at the end of the acquisition; against additive noise this is provably the safe choice.

Dynamic media break this instinct in a specific, instructive way. A time-varying channel multiplies the bucket record by a slowly drifting gain: the $n$-th measurement is $R_n = a_n B_n$, with $B_n$ the object-dependent bucket coefficient and $a_n > 0$ an unknown per-frame gain whose log-gain $\ell_n = \log a_n$ is confined to a slow-drift class $S$ of dimension $p$. Ghost imaging through dynamic and complex scattering media [7] shows a self-calibrating scheme built on *random* illumination with supervised correction of dynamic scaling factors succeeding where ordered scans fail, and optical wireless transmission around corners [8] relies on the same random-pattern robustness. Read naively, this is a puzzle: the ordered Hadamard scan is the better-conditioned instrument, yet randomness — usually a concession, not a feature — wins; it is desirable to determine when, and by what mechanism, a drifting gain becomes blindly separable at all.

The resolution is that under gain drift the ordering itself becomes the liability. An ordered scan sorts the object's transform coefficients into a smooth sequence in acquisition time, so the object acquires a *temporal signature*: a slow gain trace and a slow coefficient trace cast the same shadow on the one-dimensional bucket record, and no estimator can tell them apart from that record alone. Hadamard is not defeated by randomness; it is defeated by chronology. This confounding is not an artifact of any particular correction algorithm but a genuine algebraic ambiguity — for a square unconstrained design, *any* invertible pattern matrix can absorb a nonconstant gain into a perfectly valid alternative object (Theorem A). Conversely, random carriers escape it statistically: when the pattern sequence makes the bucket carrier's distribution stationary in time, slow gain becomes the only non-stationary component of the record, and the relative gain becomes identifiable up to the unavoidable global scale (Theorem B). Self-calibration is thus a property one can *design into acquisition time*, with price, rates, and failure boundaries computable in advance.

We develop that observation into a quantitative identifiability theory; the contributions are:

1) **A three-notion identifiability framework** for the bilinear model $R_n = a_n B_n$, separating exact-algebraic identifiability, statistical (asymptotic) identifiability, and estimation conditioning. Within it, the square-design obstruction (Theorem A) and the tall-design thresholds (Theorem A′) are proved for every fixed drift class $S$ containing the constants: generic exact identifiability at $N \geq K + p$ for almost every design–object pair, uniform identifiability over every nonvanishing-carrier object at $N \geq 2K+p-1$, and a generic local threshold at $N = K + p - 1$ that is an if-and-only-if for $K \geq 2$, with genuine positive-dimensional alias families below it (Secs. 4, 9). This is distinct from generic bilinear-injectivity results [13], [14], [15], [16], [17]: the operator here is a diagonal-row map,

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and the anchor is temporal, not structural.

2) **The stationarity anchor (Theorem B)** and its consistency condition ($\star$): random carriers pin the relative gain statistically, with a windowed log-domain estimator achieving the minimax rate over the drift class; a separate *oracle-known-carrier* (calibrated soft-log) result covers the photon-limited case, scaling as $1/(W\bar{\lambda})$ in the locally-flat-gain regime — the blind high-count and known-carrier low-photon theorems are distinct and not merged.
3) **A synthesis, not a trade-off:** randomized Hadamard (SRHT) designs retain orthogonal conditioning while randomizing the temporal carrier — stationary/exchangeable under Walsh-flatness of $T^2$ or a successful random permutation of a sufficiently spread object — with an exact Walsh-domain iff whitening condition separating the roles of sign flips and permutations.
4) **An exact finite-noise reconstruction bridge** (Theorem 1) expressing relative MSE for any linear reconstructor through a leverage functional $B_L$, covering residual gain, read noise, and Poisson noise without Gaussian approximation under explicit exogenous-gain hypotheses, specializing to orthogonal, pairwise-Hadamard, and random/DGI pipelines [18], [19] (DGI constants measured per implementation).
5) **A design chart with a finite-$N$ flip boundary** $\rho^*$ in $\rho_{\mathrm{pair}}$ (Prop 3), giving a pre-acquisition go/no-go rule (Fig. 4); the boundary's gain-known skeleton is validated with no free parameters, and its corrected blind-arm form predicts — and the data confirm — no flip for mean-subtracted blind DGI pipelines on the simulated grid.

All results are theoretical, supported by numerical evidence; no hardware validation is reported. Identifiability near $N = K + p$ is a uniqueness statement, not a conditioning guarantee, and random square designs remain algebraically non-identifiable — their advantage is statistical. The organizing idea is the distilled paradox: one makes the gain observable not by measuring it directly, but by making the object stop pretending to be time.

Figure 1 maps ordered-Hadamard, iid-random, tall-random, and SRHT designs on the resulting conditioning-versus-identifiability axes; Sec. 10 collects the design rules, scope, and outlook.

## II. Problem Setup and Three Notions of Identifiability

In a single-pixel (bucket-detection) acquisition of $N$ frames, frame $n$ illuminates a static object $T \in \mathbb{R}^K$ with a programmed pattern $m_n \in \mathbb{R}^K$, producing the ideal bucket value $B_n = m_n^\mathsf{T} T$ — the *object carrier*. The record is corrupted by a slowly drifting multiplicative gain (source power, detector responsivity, coupling):

$$R_n \;=\; a_n B_n, \qquad a_n = e^{\ell_n} > 0,$$

where $a_n$ is the unknown per-frame gain and the log-gain trajectory $\ell = (\ell_1, \ldots, \ell_N)$ is confined to a *drift class* $S \subset \mathbb{R}^N$, $p = \dim S$, always containing the constants. The constant direction is a pure gauge: $(a, T) \mapsto (c\,a,\ T/c)$ leaves

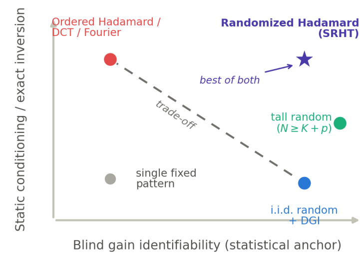


Fig. 1. Two-axis map of inversion conditioning against blind identifiability. Ordered orthogonal scans (Hadamard/DCT/Fourier) are well-conditioned yet blindly non-identifiable; i.i.d.-random (DGI) acquisition buys the statistical anchor at the cost of conditioning; a single fixed pattern (grey) has neither; tall random designs ($N \geq K + p$) restore exact identifiability modulo scale for almost every design–object pair (Theorem A′(ii)); SRHT (star) retains orthogonal conditioning while randomizing the temporal carrier. The dashed diagonal marks the apparent trade-off the SRHT synthesis dissolves (Secs. 4 and 5).

every $R_n$ invariant, so all statements are up to a global scale. For physically slow drift, $S$ is a low-pass (Fourier) space of normalized bandwidth $\rho_{\mathrm{bw}}$, $p \approx \rho_{\mathrm{bw}} N$ (not to be confused with the adjacent-pair decorrelation $\rho_{\mathrm{pair}}$ of Sec. 8). The blind problem — recover $(a, T)$ from $\{R_n\}$ alone — is bilinear, and asking whether it is "solvable" turns out to be three distinct questions; separating them is itself a contribution of this paper.

**Conventions (fixed throughout).** $K$ — pixels; $T \in \mathbb{R}^K$ the object, nonnegative unless stated; $N$ — frames; $S$ — linear log-gain drift class, **always containing the constants**, $p = \dim S$. **All** identifiability statements are modulo the gauge $(a, T) \mapsto (c\,a, T/c)$, fixed by centering the log-gain; reconstruction metrics use least-squares scale alignment (Sec. 8). *Local-differential*, *generic-exact*, and *uniform-exact* identifiability are distinct properties with distinct thresholds (Theorem A′); none implies conditioning. $\rho_{\mathrm{bw}}$ — normalized gain bandwidth, $p \approx \rho_{\mathrm{bw}} N$ (Sec. 4); $\rho_{\mathrm{pair}}$ — adjacent-pair gain decorrelation (Sec. 8); related but not equal. The object functionals $K_{\mathrm{eff}}$, $K_\infty$, $K_4$ and pipeline constants $C_0$, $D_H$, $B_L$ are defined at first use (Prop. 1, Secs. 6–8); a consolidated notation table appears in Supplementary Material S5.

Pattern entries $m_n$ may be signed (Hadamard) or nonnegative (speckle); where signed coefficients appear as carriers, the Sec. 5 bucket conventions specify their physical realization.

**Algebraic identifiability.** Does the noiseless record determine $(a, T)$ up to gauge, or does some other pair with $\ell' - \ell \in S$ nonconstant reproduce the data exactly — a finite-sample, deterministic question about collisions $MT = D_s M T'$? Section 4 shows that every square invertible design, however well-conditioned, fails it, and that overdetermination restores it generically at $N \geq K + p$.

**Statistical identifiability.** Under random pattern ensembles a weaker but operationally decisive property becomes available: if the carrier $B_n$ is *stationary* — its distribution does not depend on $n$ — then any slow modulation of local averages over windows $W$ is attributable to the gain alone, and the relative gain is identifiable in distribution even where exact algebraic separation is impossible. This stationarity anchor (Sec. 5, Theorem B) is the mechanism the field's empirical successes have been implicitly exploiting.

**Estimation conditioning.** Uniqueness is not accuracy: even when one of the first two notions holds, the finite-noise error of a concrete estimator is governed by conditioning quantities — the window $W$, the residual-gain variance $v$, the leverage $B_L$ — quantified in Secs. 6 and 8. Identifiability at the threshold $N = K + p$ says nothing about stability there.

Much of the confusion surrounding blind gain correction stems from conflating these three notions: judging a scheme by the visual quality of a blindly corrected reconstruction entangles algebraic ambiguity, the statistical anchor, and estimator conditioning into a single scalar — three failures with different remedies. Figure 1 disentangles them as a two-axis map — inversion conditioning against blind identifiability — on which ordered Hadamard, i.i.d.-random (DGI), tall random, and SRHT acquisitions occupy visibly different corners (Secs. 4 and 5).

## III. Relation to prior work

Gain–object separation is a bilinear inverse problem: lifting the unknowns to the rank-one array $X = aT^\top$, each bucket $R_n = a_n B_n$ with $B_n = m_n^\top T$ is a bilinear form in the object $T$ and the gain trace $a$. This places the question inside a mature literature on bilinear identifiability; we state what it settles and what it leaves open for the diagonal-gain geometry.

The sharpest injectivity counts for lifted bilinear maps are Kech–Krahmer's optimal conditions for generic identifiability [13], requiring on the order of $2(n_1 + n_2) - 4$ generic rank-one measurements. These counts do not transfer here: the operator $L_M(X)_n = a_n\, m_n^\top T$ is a *diagonal-row* map, not a generic rank-one measurement, and the relevant question is the intersection geometry of the object subspace with its gain-modulated copy, $U \cap D_s U$. Choudhary–Mitra's scaling laws [16] are the closest in spirit to our Theorem A: their rank-two nullspace obstruction is exactly the mechanism by which a square unconstrained design absorbs a nonconstant gain into a redefined object. Li–Lee–Bresler's unified framework [14] supplies the subspace- and sparsity-constrained thresholds that our tall-design counts ($N \geq K + p$ generic, $N \geq 2K + p - 1$ uniform) specialize to for the diagonal-gain model. Ahmed–Recht–Romberg [15] establish convex recovery for blind deconvolution under subspace priors, and Kueng–Rauhut–Terstiege [17] give low-rank-from-rank-one-measurement guarantees; both are the algorithmic backdrop for our purely algebraic and statistical criteria.

On the imaging side, single-pixel practice has long weighed ordered Hadamard against random speckle for conditioning and noise, but without an identifiability account of slow multiplicative drift. Learned correction of dynamic scaling factors [7] and learned encoding for optical wireless links [8] address the same nuisance empirically; our analysis explains why their *static-correction ceiling* — the error floor left by any time-independent correction applied to a drifting record — is the honest bound, and when a drifting gain becomes blindly separable at all. Blind self-calibration of sensor gains and phases has a long history outside imaging — eigenstructure methods [20], blind sensor-network calibration [21], biconvex compressive self-calibration [22] — and Balzano–Nowak similarly find gains recoverable under mild oversampling, which our threshold $N \geq K + p$ makes exact for the diagonal bilinear model. On the acquisition-design side, pink-noise speckle [23] and sparse Hadamard sequencing [24] optimize the pattern set directly but target additive-noise robustness, not multiplicative-drift identifiability.

Against this background the contribution is specific: we do not add to generic bilinear injectivity, settled above. New is the temporal-statistical mechanism — relative-gain identifiability secured by making the carrier stationary (Theorem B), the associated nonparametric calibration rate, the ordered-orthogonal confounding failure mode it repairs, the SRHT synthesis, and the finite-noise bridge that renders the phase diagram quantitative. The novelty is the stationarity anchor, not bilinear injectivity per se.

## IV. The algebraic obstruction and when oversampling cures it

The paradox of Sec. 1 is not an estimator artifact. Stack $N = K$ frames of the noiseless record $R_n = a_n B_n$ into an invertible square design $M$: the multiplicative drift is absorbed *exactly* into a relabelled object.

**Theorem A (square non-identifiability).** *For any invertible $M \in \mathbb{R}^{K\times K}$ and any gain $a = (a_n)$ nonconstant on the support of the carrier $MT$, there exists a distinct object $T' \neq cT$ producing the identical bucket record:* $\operatorname{diag}(a)MT = MT'$ *with* $T' = M^{-1}\operatorname{diag}(a)MT$. *Gain–object separation is unidentifiable up to the global-scale gauge $a \mapsto ca$, $T \mapsto T/c$. (Proof: Appendix A.)*

The object is a barcode printed onto time and the drift a smudge across it; when the barcode carries slow structure, the smudge can be *reread as a different barcode* rather than dismissed as blur — a genuine orbit of the bilinear map, for *every* invertible design. Two corollaries sharpen the trap (full statements and proofs: Appendix A). *Corollary 1 (slowness does not help):* restricting $\ell \in S$ to a slow drift class does not restore identifiability at $N = K$ — for any nonconstant $s \in S$ the alternative pair $\big(\ell - s,\ T' = M^{-1}\operatorname{diag}(e^s)MT\big)$ explains the record exactly, and $\ell - s$ is again an admissible slow drift because $S$ is linear: a slowly varying gain masquerades as a slowly varying object. *Corollary 2 (positivity is not enough; known support zeros can be):* nonnegativity of $T$ alone does not break the orbit, but $K_0 \geq p - 1$ *known* generic support zeros — one per nonconstant drift dimension — restore *local* identifiability via a rank condition on the constrained collision map; a local rank criterion, not a global guarantee.

The first escape is overdetermination. For a tall design $M \in \mathbb{R}^{N\times K}$, $N > K$, a collision $\operatorname{diag}(e^{s'})MT' = \operatorname{diag}(e^s)MT$ pits $K + p$ unknowns against $N$ equations, suggesting uniqueness once $N \geq K + p$; the following theorem turns this count into proved sufficiency thresholds, for every fixed drift class $S \ni \mathbf{1}$, at the price of almost-everywhere quantifiers with explicit Lebesgue-null exceptional sets.

**Theorem A′ (proved tall-design identifiability results).** *Let $K \geq 1$, $N \geq K$, and let $S \leq \mathbb{R}^N$ be a fixed, known linear subspace of dimension $p \geq 2$ with $\mathbf{1} \in S$. For $M \in \mathbb{R}^{N\times K}$ define $\Phi_M(T, s) = D_{e^s}MT$ and $\mathcal{U}_M = \{T \in \mathbb{R}^K : (MT)_n \neq$*

$0$ *for every* $n\}$. *Parameters are identified modulo the positive global-scale gauge* $(T,s) \sim (cT,\ s - \log c\,\mathbf{1})$, $c > 0$.

(i) (Generic local threshold and genuine failure below it, $K \geq 2$.) *There is a Lebesgue-null set* $E_{\mathrm{loc}} \subset \mathbb{R}^{N\times K} \times \mathbb{R}^K$ *such that for every* $(M,T) \notin E_{\mathrm{loc}}$ *with* $T \in \mathcal{U}_M$ *and every* $s \in S$: $\operatorname{rank} D\Phi_M(T,s) = \min\{N, K+p-1\}$. *Consequently* $(T,s)$ *is locally identifiable modulo gauge iff* $N \geq K+p-1$. *If* $N \leq K+p-2$ *then a gauge-normalized neighborhood of* $(T,s)$ *contains a real-analytic manifold of exact aliases of dimension* $K+p-N-1 \geq 1$, *and every other sufficiently nearby point of it is gauge-inequivalent.*

(ii) (Generic exact-identifiability sufficiency.) *If* $N \geq K+p$, *there is a Lebesgue-null set* $E_{\mathrm{gen}} \subset \mathbb{R}^{N\times K} \times \mathbb{R}^K$ *such that for every* $(M,T) \notin E_{\mathrm{gen}}$ *and every* $s \in S$: $D_{e^{s'}}MT' = D_{e^s}MT$ *with* $(T',s') \in \mathbb{R}^K \times S$ *implies* $T' = cT$, $s' = s - \log c\,\mathbf{1}$ *for a unique* $c > 0$. *Exact identifiability holds simultaneously for every possible true drift* $s \in S$. *This statement is **not** uniform over all objects for one fixed* $M$.

(iii) (Uniform exact-identifiability sufficiency on the nonvanishing-carrier set.) *If* $N \geq 2K+p-1$, *there is a Lebesgue-null set* $E_{\mathrm{unif}} \subset \mathbb{R}^{N\times K}$ *such that for every* $M \notin E_{\mathrm{unif}}$, *every* $T \in \mathcal{U}_M$, *every* $s \in S$, *and every* $(T',s') \in \mathbb{R}^K \times S$: $D_{e^{s'}}MT' = D_{e^s}MT$ *implies* $T' = cT$, $s' = s - \log c\,\mathbf{1}$ *for a unique* $c > 0$. *The same generic design works simultaneously for all nonvanishing-carrier objects and all true drifts in* $S$.

(iv) (Degenerate one-dimensional object stratum.) *If* $K = 1$ *then for every* $M$ *and every* $T \in \mathcal{U}_M$ *and every* $s \in S$, *the pair* $(T,s)$ *is globally identifiable modulo gauge, for every admissible* $N$. *Neither* $N \geq K+p$ *nor* $N \geq 2K+p-1$ *is necessary when* $K = 1$.

*(Proof: Appendix B.)*

**Sharpness status.** Parts (ii) and (iii) are sufficient conditions, not proved if-and-only-if thresholds. For $K \geq 2$, the below-wall part of (i) proves generic exact failure whenever $N \leq K+p-2$. The remaining necessity assertion for the $K+p$ bound is the boundary case $N = K+p-1$: dimension counting predicts nonlocal aliases there, but dominance of the bad-pair projection has not been proved. Likewise, dimension counting predicts that a generic design fails uniform identifiability for some object when $N \leq 2K+p-2$, but this necessity direction has not been proved in the intermediate regime. Thus $K+p$ and $2K+p-1$ are conjecturally sharp for $K \geq 2$ under generic/nondegenerate drift geometry, not universal sharp thresholds for every fixed $S$. The $K = 1$ stratum explicitly rules out any universal necessity claim.

The proof is a parametric-transversality argument (Appendix B): a rowwise perturbation of $M$ acts as a global witness on a collision incidence manifold, and no genericity of the drift family is invoked, so the statements hold verbatim for the physical Fourier low-pass class $S_{\mathrm{LP}}$ and any other fixed drift convention containing the constants. Empirically, the $K = 64$ and $K = 128$ threshold scans of Sec. 9 (Fig. 6) show an exact $0\% \to 100\%$ rank transition at $N = K+p-1$ across all five drift dimensions at both object sizes (3250 cells, no exceptions) — a direct verification of the generic local rank wall of part (i). Part (ii) asserts *generic exact finite-sample identifiability*, not a conditioning bound and not an algorithmic recovery guarantee; stability near the thresholds is the separate third notion of Sec. 2, quantified in Secs. 6 and 8.

The picture is a dichotomy. When $N \geq K+p$, overdetermination *alone* separates gain from object for almost every design–object pair — no statistical mechanism is logically needed, though randomization still buys stable conditioning (Fig. 1). When $K < N \leq K+p-2$, generic local and exact identifiability fail through positive-dimensional nearby alias families; at $N = K+p-1$, local identifiability holds generically, while failure of global exact identifiability remains a dimension-count prediction. In that under-resourced window the stationarity anchor of Sec. 5 (or a support/sparsity prior) becomes *required*, not merely convenient. The regimes meet through the drift bandwidth: a gain occupying a fraction $\rho_{\mathrm{bw}}$ of the frame band spans $p \approx \rho_{\mathrm{bw}}N$ low-pass modes, so $N \geq K+p$ reads as $N \gtrsim K/(1-\rho_{\mathrm{bw}})$. Slow drift ($\rho_{\mathrm{bw}} \ll 1$) is cured by modest oversampling; as $\rho_{\mathrm{bw}} \to 1$, $p \to N$ and blind algebraic separation is impossible for *any* $N$, forcing the statistical route.

## V. Statistical identifiability: the stationarity anchor

Section 4 bought identifiability with surplus rows; this section develops a second, logically distinct escape that operates even at fixed $N$ and explains *why* randomized acquisition self-calibrates while ordered acquisition does not. A drifting gain is a boat carried by a slow current, and the bucket record is our only view of the water: if the waves change character with position — as the ordered Hadamard coefficient sequence does — we cannot tell the current from the coastline; if the waves are *statistically identical everywhere*, any slow modulation of the record has exactly one explanation. Local averaging then pins the *relative* drift, the absolute scale remaining gauged away. This intuition, and nothing stronger, is what the results below formalize.

The log-domain record can be described by $Y_n = \log R_n = \ell_n + \log B_n$. The anchor requirement is a condition on the carrier alone:

**Condition ($\star$) (stationary carrier, quantitative form).** There is a scalar $m_T$ — depending on the object but not on time, and absorbed by the global-scale gauge — such that the windowed carrier averages over windows $W_n$ of length $W$ satisfy

$$\max_n \left| W^{-1} \sum_{k\in W_n} \log B_k \ - \ m_T \right| \ \longrightarrow\ 0$$

in probability as $W \to \infty,\ W/N \to 0$.

The condition presupposes a positive carrier on the analyzed record (a positivity/offset margin, or the Sec. 6 soft-log in the photon-counting case).

The criterion is deliberately *uniform* (a maximum over all window locations); the pointwise-to-uniform upgrade typically costs a $\log N$ refinement (the $\delta \mapsto \delta/N$ step of Appendix C).

I.i.d. random illumination satisfies ($\star$) by construction. For a finite random row permutation or a single fixed SRHT draw it is *not* automatic: a permuted record is exchangeable — anchoring windowed averages in distribution — but a particular draw can still fail for a spectrally concentrated object (the quantitative pixel-permutation version is the Sec. 7 permutation bound; pixel-level and acquisition-order randomization are distinct — Appendix E.6), and the permuted-arm rejections in Fig. 2 are its visible trace. For the i.i.d. case:

**Proposition 1 (random-basis carrier).** *For i.i.d. random patterns with pixel mean $\mu_I$ and standard deviation $\sigma_I$, the carrier $\{B_n\}$ is white and strictly stationary, and the object enters its first two moments only through the scalars $S_1 = \sum_j T_j$ and $S_2 = \sum_j T_j^2$: $\mathbb{E}B_n = \mu_I S_1$, $\operatorname{Var} B_n = \sigma_I^2 S_2$, hence $\mathrm{CV}_B = (\sigma_I/\mu_I)/\sqrt{K_{\mathrm{eff}}}$ with $K_{\mathrm{eff}} = S_1^2/S_2$. Gaussian and log-domain approximations additionally require the Lindeberg-type spikiness condition $K_\infty = S_2/\|T\|_\infty^2 \to \infty$; the reduction of the object to two scalars holds at this Gaussian-approximation level, not exactly. Higher cumulants and the log-domain mean can depend on the object's shape beyond $(S_1, S_2)$, but for an i.i.d. carrier every such term is time-independent and shifts only the gauge scalar $m_T$; it does not affect relative-gain recovery. (Proof: Appendix C.)*

Figure 2 shows the two carriers side by side: the random trace is a flat, object-independent noise band, whereas the ordered Hadamard trace *is* the object's coefficient envelope — the object masquerades as a temporal trend, precisely the degree of freedom a slow gain occupies, and ($\star$) fails. The condition's logical status:

**Proposition 2 (($\star$) is an estimator criterion, not a universal one).** *Condition ($\star$) is necessary and sufficient for consistency of the blind windowed (stationarity-anchor) gain correction. It is* not *an if-and-only-if condition for identifiability by all mechanisms: the overdetermined designs of Sec. 4 identify algebraically — for almost every design–object pair at $N \geq K + p$, Theorem A′(ii) — without any stationarity, and conversely ($\star$) confers no exact algebraic identifiability. (Proof: Appendix C.)*

**Bucket conventions.** Three bucket-signal conventions coexist (full statement: Supplementary Material S5): (i) *signed coefficients* — $B_n$ can vanish or change sign, so $\log B_n$ is undefined; (ii) *physical nonnegative offset* — $B_n \geq B_{\min} > 0$; Theorem B and the Sec. 6 estimators live here (assumption (iv) below); (iii) $\pm$ *pairwise* — complementary pairs differenced, drift entering through $\rho_{\mathrm{pair}}$ (Sec. 8). The raw signed arms of Sec. 9 follow (i) and are *diagnostic only*.

**Theorem B (statistical relative-gain identifiability).** *Assume the log-domain model*

$$Y_n = \log R_n = \ell_n + m_T + z_n,$$

*where (i) $m_T = \mathbb{E}\log B_n$ is a scalar depending on the object but not on time; (ii) the carrier fluctuations $\{z_n\}$ are centered, strictly stationary, and weakly dependent with summable autocovariances, $\sigma_{\mathrm{abs}}^2 = \sum_h |\gamma(h)| < \infty$ where $\gamma(h) = \mathrm{Cov}(z_0, z_h)$ ($\beta$-mixing with summable coefficients is sufficient) — the* signed long-run variance $\sigma_{\mathrm{LR}}^2 = \gamma(0)+2\sum_{h\geq 1}\gamma(h)$ ($\leq \sigma_{\mathrm{abs}}^2$) *then also exists and is the asymptotic variance of window means; the two constants play distinct roles below; (iii) the log-gain $\ell \in S$ lies in a normalized-time Hölder-$\beta$ class, $|\ell_j - \ell_k| \leq L\,(|j-k|/N)^\beta$; and (iv) $R_n > 0$ on the analyzed record so the logarithm is defined (positivity/offset margin; the soft-log of Sec. 6 covers the photon-counting case). Then, for windows satisfying **both** $W \to \infty$ **and** $W/N \to 0$, windowed averages of $Y_n$ estimate $\ell_n + m_T$ consistently, with non-asymptotic stochastic error $O_P(\sigma_{\mathrm{abs}}/\sqrt{W})$ — the asymptotic variance of the window mean being $\sigma_{\mathrm{LR}}^2/W$ — and windowed drift bias $O(L\,(W/N)^\beta)$; consequently the* centered *log-gain $\ell_n - \bar{\ell}$ — the relative gain, up to one global scale — is identifiable and consistently estimable. The scalar $m_T$, and with it the absolute gauge of $\ell$, is not recoverable. (Proof: Appendix C.)*

Theorem B must be read with its scope intact: it is an asymptotic, high-probability statement about *relative*-gain recovery up to the gauge — never exact finite-sample identifiability of $(a, T)$ — so it does not contradict Theorem A: a random square design remains algebraically non-identifiable; randomness buys statistical separability, a different currency. This also distinguishes the mechanism from the bilinear-inverse literature [14], [13], [16]: the anchor is a property of the carrier's *temporal distribution*, not of subspace injectivity.

The anchor also explains why the self-calibrating scaling-factor correction of Ref. [7] can work at all: under Proposition 1's Gaussian approximation the bucket record is a slow trend plus stationary residuals, and an SCGI-style Gaussian-likelihood fit of that trend is a maximum-likelihood realization of the windowed anchor estimator, inheriting Theorem B's consistency guarantee (and its gauge).

Figure 3 is the core numerical evidence: under random illumination the blind gain error $\|\hat{a} - a\|/\|a\|$ tracks $\mathrm{CV}_B/\sqrt{W}$ object by object — to leading Gaussian/log-variance order the object enters only through $K_{\mathrm{eff}}$ — and normalized by that floor the curves collapse to $\pm 5$–$7\%$; under ordered Hadamard the curves scatter by object and do not decay with $W$ — the object envelope is a bias no amount of averaging removes. The window $W$ carries a bias–variance trade-off (drift curvature versus $\mathrm{CV}_B^2/W$), resolved with minimax rate in Sec. 6.

## VI. Estimation rate, lower bounds, and low-photon robustness

Section 5 establishes that relative gain is *identifiable* from a stationary carrier; here we show it is recovered *efficiently and predictably*. The baseline estimator is the plain windowed log of Theorem B: average $Y_n = \log R_n$ over a window $W$ of frames sharing a slowly varying gain. For photon-counting records we use its robust generalization, the windowed *soft-log*: average $\psi_\alpha(c) = \log(c+\alpha)$ and invert the per-window calibration curve $M_n(\theta) = \sum_k w_{nk}\, m_\alpha(\Lambda_0 e^\theta b_k + d)$ built from the *known* carriers $b_k$, with $m_\alpha(\lambda) = \mathbb{E}[\psi_\alpha(C)]$, $C \sim \mathrm{Pois}(\lambda)$ — an *oracle-known-carrier* estimator (the homogeneous shortcut $m_\alpha^{-1}(\mathrm{mean}_W\, \psi_\alpha)$ is a mean-Poisson proxy, exact only for constant carriers); as $\alpha \to 0$ at high $\bar{\lambda}$ it reduces to the plain windowed log ($\beta$ is reserved for Hölder smoothness, $\alpha$ for the soft-log offset). The window trades bias against variance in the usual nonparametric way: wider $W$ suppresses variance as $O(1/W)$ but incurs a bias set by the smoothness $\beta$ and the drift Lipschitz constant $L_a$ ($= L\,N^{-\beta}$, the per-frame form of Theorem B's normalized-time constant).

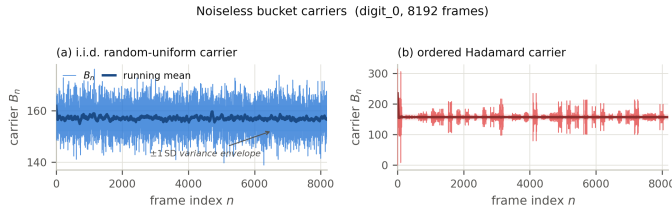


Fig. 2. Carrier stationarity: noiseless bucket carriers $B_n$ (thin), running means (bold), and the $\pm 1$ SD envelope for one object (digit_0; 8192 frames, $64 \times 64$), before any gain is applied. (a) The i.i.d. random-uniform carrier is a flat, object-independent noise band; (b) the ordered Hadamard trace *is* the object's coefficient envelope — a deterministic, non-stationary temporal signature. Rejection counts and the Parseval invariant: Sec. 9.1.


(a) (b)

Fig. 3. Blind gain identifiability (seven frame-matched arms, 2048 frames each; $32 \times 32$ objects, $K = 1024$; medians over 20 seeds $\times$ 10 objects at $\rho_{\text{pair}} = 10^{-3}$, $s = 0.1$). (a) Blind gain error $\|\hat{a} - a\|/\|a\|$ versus window $W$ under the ratio AGC (solid, filled) and the Theorem-B log estimator on the $R > 0$ record (dashed, open); raw signed-Walsh arms are diagnostic-only. Error bars: two-way clustered 95% CIs of the median best-window error. (b) Errors normalized by the theory floor $\sqrt{\mathrm{CV}_B^2/W}$ collapse across the ten objects. Quantitative values: Sec. 9.2 and Table III.

**Theorem C (windowed rate).** *For log-gain in a Hölder-$\beta$ class with constant $L_a$, the bias-optimized windowed estimator attains*

$$\mathrm{MSE}^* \;\le\; C\,\underline{\kappa}^{-2}\left[\kappa_{\max} L_a\right]^{2/(2\beta+1)} \bar{\sigma}_\alpha^{4\beta/(2\beta+1)},$$

*the standard nonparametric rate in the effective per-frame noise $\bar{\sigma}_\alpha$ ($\underline{\kappa}$: the operating-range window-sensitivity infimum; Appendix D.4). (Proof: Appendix D.)*

This rate is optimal: an Assouad two-point construction [25] over the same smoothness class (a van Trees [26] / Pinsker [27] argument in the Sobolev variant) yields a matching lower bound, so no estimator improves on Theorem C by more than constants (photon-counting case: the direct Poisson Le Cam bound of Appendix D.4).

**Theorem D (minimax optimality).** *Over the Hölder-$\beta$ drift class the windowed estimator is minimax-rate optimal: the lower bound matches Theorem C at the level of rates. Sharp minimax constants remain open. (Proof: Appendix D.)*

The floor beneath both bounds is Fisher geometry: in the gauge quotient (global scale unobservable, Sec. 5) the relevant object is the *quotient Fisher information*, and it is singular exactly along the null direction of Theorem A. Non-identifiability is not poor conditioning; it is a genuine flat direction of the likelihood — the exact collision curve $\ell_u = \ell + u$, $T_u = M^{-1}\operatorname{diag}(e^{-u})MT$ of Theorem A made infinitesimal (tall designs need the range condition of Appendix D.3; constant $u$ is the gauge).

Which design constant controls what is a spikiness question, and three distinct functionals appear: $K_{\text{eff}}$ governs excitation and the pairwise-Hadamard residual-gain budget (Prop 1); $K_\infty$ is the Lindeberg parameter gating the Gaussian/log-domain approximations and windowed-energy concentration; $K_4$ enters the permutation-whitening concentration guarantee (Sec. 7, via $\min(\epsilon^2 K_4, \epsilon K_\infty) \ge C\log(K/\delta)$). Conflating them mislabels the bottleneck: a design flat in $K_{\text{eff}}$ can still be spiky in $K_\infty$.

Low photon counts do not break the log domain: the Poisson derivative identity $\frac{d}{d\lambda}\mathbb{E}f(C) = \mathbb{E}[f(C{+}1) - f(C)]$ gives sensitivity $\kappa_\alpha(\lambda) \to 1$ at high $\bar{\lambda}$ and $\kappa_\alpha(\lambda) \to \lambda\log(1 + 1/\alpha)$ as $\bar{\lambda} \to 0$, and the offset $\alpha$ absorbs empty bins so no frame contributes $\log 0$. Through Theorem C the photon-starved variance factor obeys the general window law of Appendix D.4 (actual intensities in the numerator, anchor intensities in the sensitivity denominator), simplifying to $\sim 1/(W\bar{\lambda})$ only under *local gain flatness* — where it matches the direct Poisson lower bound at rate level, since at $d = 0$ the Poisson information for a known-carrier log-intensity shift is $\lambda$ itself: zero photons carry zero Fisher information; they cut precision but never diverge (Fig. 5). The reference curve plotted there, $\mathrm{mean}_n[1/I_n]$, is an *oracle common-shift diagnostic*, not a nuisance-path quotient Cramér–Rao bound.

The caveat, stated honestly: without calibration inversion, a plain windowed average of $\psi_\alpha$ — the *uncalibrated proxy* — becomes shrinkage-bias-dominated at $\bar{\lambda} \lesssim 1$: it is pulled toward $\log\alpha$, and its MSE sits *below* the oracle common-shift diagnostic because bias is present (MSE below $1/I_A$ is compatible with shrinkage bias; MSE above it does not certify unbiasedness — Appendix D.4). The calibrated estimator of Theorem C removes this shrinkage and tracks the $1/(W\bar{\lambda})$ law without falling below the diagnostic on the tested grid, giving up precisely the bias-bought low-count MSE discount of the proxy (and of the Anscombe transform [28]) — quantified in Sec. 9 (Fig. 5).

## VII. Basis design: randomized orthogonal acquisition (SRHT)

Must one surrender orthogonal conditioning to buy the anchor's random illumination? No. The randomized Hadamard transform — an orthogonal Hadamard carrier composed with a random sign flip and a random pixel permutation, the SRHT of the sketching literature — keeps a full-rank, well-conditioned forward map while randomizing the *temporal handwriting* of the object: one shuffles the deck without bending the cards. We use the *full* $N = K$ randomized transform throughout; "subsampled" is reserved for the $N < K$ case, which raises a bias/prior question outside our scope.

The mechanism is exact. Writing the per-coefficient excitation as $Z_g$ under a random diagonal sign $D$, the residual gain–object coupling is governed entirely by the Walsh spectrum of the squared object. With $w = T^2$,

$$\mathrm{Cov}_D(Z_g, Z_h) = \hat{w}(g + h),$$

the Walsh transform of $w$ at the dyadic index sum. Sign randomization alone therefore whitens the excitation **iff** $w$ is Walsh-flat ($\hat{w}(k)$ negligible for all $k \neq 0$) — a condition on the object, not the design: a structured $T$ concentrates $\hat{w}$ off the origin and sign flips cannot remove it.

The random permutation supplies the missing genericity, acting as a probabilistic Walsh-flattener: permuting pixels spreads the mass of $\hat{w}$ across the spectrum, driving the off-origin covariance below tolerance $\varepsilon$ with probability $1-\delta$ once the object is spectrally spread enough,

$$\min\bigl(\varepsilon^2 K_4,\ \varepsilon K_\infty\bigr) \ \geq\ C \log(K/\delta),$$

with $K_4$, $K_\infty$ the Sec. 6 spread functionals — both diverge as the object's energy delocalizes and collapse when a few pixels dominate; the bound quantifies how much delocalization the permutation needs.

Permutation is a genuine second ingredient. Under permutation alone,

$$\mathrm{Var}_P\, Z_g = \frac{K S_2 - S_1^2}{K-1} = S_2\, \frac{K - K_{\mathrm{eff}}}{K-1},$$

so the residual carries an $O(1/K_{\mathrm{eff}})$ *upper* bound on variance but no matching lower bound: a flat object produces zero excitation, so permutation cannot guarantee two-sided control coefficient by coefficient, and the whitening claim is one-sided unless the sign flip and the spectral-spread condition are both in force. A row-order shuffle of the acquisition sequence is a third, weaker operation: it makes the temporal carrier exchangeable — enough for the windowed anchor of Sec. 5 to hold in distribution — but does not guarantee two-sided per-coefficient excitation. The ablation of Supplementary Material S1 (Fig. 8) probes the acquisition-time operations (none / row-order permutation / sign / both); its permutation arms reorder acquisition time ($P_{\mathrm{row}}$) and do *not* realize the pixel permutation $P_{\mathrm{col}}$ of the whitening analysis above, whose factorized attribution — including the $P_{\mathrm{col}}$-only and $P_{\mathrm{col}}+D$ arms in natural acquisition order — is reported in Sec. 9.6 and Appendix E.6.

**Design rule (blind slow gain).** (i) Do *not* acquire in ordered Hadamard when the nuisance is a blind, slowly drifting gain — the ordering hands the object a temporal signature (Sec. 4). (ii) Randomize the carrier: apply the SRHT sign flip *and* permutation. (iii) Verify Walsh-flatness of the realized (permuted) $T^2$, or rely on the permutation bound above with $K_4, K_\infty$ estimated from the object class. (iv) Keep a positivity/offset margin so the log or soft-log stage of Sec. 6 stays away from its singularity. (v) Choose the averaging window $W$ from the bias–variance trade of Sec. 6, not from conditioning.

The rule prescribes *both* randomizations because that configuration has a two-sided theorem guarantee under the stated spectral-spread condition; the ablation (Fig. 8) reports the different, empirical fact that either randomization alone already recovered essentially the full advantage in the tested panel. The two statements must not be merged: the theorem guarantees whitening under its hypotheses, the ablation shows that any tested way of breaking the chronology sufficed here; we do not promote the panel result to a universal claim.

The synthesis: orthogonality and self-calibration are compatible. SRHT preserves the $B_L = 1$ leverage of a full orthogonal inversion (Sec. 8) while — under Walsh-flatness of $T^2$ or a successful permutation of a sufficiently spread object — making the carrier statistically stationary in time, so the only non-stationary structure left in the record is the gain itself; the conditioning that made ordered Hadamard attractive is retained, only its chronology discarded.

(Proof of the Walsh identity and permutation bound: Appendix E.)

## VIII. From identifiability to images: the reconstruction bridge and phase diagram

Sections 4–7 answer the statistician's question — *is the gain identifiable, and at what rate?* — but the practitioner asks: *how many gray levels do I lose?* This section supplies the bridge: exact at finite $N$ and finite photon count, it reduces every acquisition strategy in this paper to three numbers readable off a single chart before any hardware is built.

The corrected bucket record is $z_n = (1 + \delta_n) b_n + \xi_n$ with $b = AT$, $\delta_n$ the *residual* multiplicative gain after whatever blind correction the pipeline applied, and $\hat{T} = Lz$ any linear reconstruction.

**Theorem 1 (master finite-noise relMSE identity).** *Assume — as part of the hypothesis, not as background convention — that (H1) $\mathbb{E}\,\xi = 0$; (H2) the residual gain $\delta$ and the additive noise $\xi$ are uncorrelated, $C_{\delta\xi} := \mathrm{Cov}(\delta, \xi) = 0$; (H3) the gain estimate $\hat{a}$ is* exogenous *with respect to the reconstructed record — computed independently of the noise realization $\xi$ (injected or synthetic residuals, an independent calibration record, or a cross-fitted estimate from held-out frames), so that (H1)–(H2) hold conditionally on the record being reconstructed; and (H4) the second moments $m_\delta$, $V_\delta$, $\Sigma_\xi$ are finite.*

*Then, without Gaussian or small-noise approximation,*

$$\frac{\mathbb{E}\|\hat{T}-T\|^2}{S_2} = \underbrace{\frac{\|(LA-I)T + L\,\mathrm{diag}(b)\,m_\delta\|^2}{S_2}}_{bias} + \underbrace{\frac{\mathrm{tr}\big(L\,\mathrm{diag}(b)\,V_\delta\,\mathrm{diag}(b)\,L^\top\big)}{S_2}}_{residual\ gain} + \underbrace{\frac{\mathrm{tr}\big(L\,\Sigma_\xi L^\top\big)}{S_2}}_{read\ +\ Poisson}.$$

*If the residual gains are independent with* $\mathrm{Var}\,\delta_n = v$, *the gain term collapses to* $v\,B_L$ *with the* ***leverage*** $B_L = (1/S_2)\sum_n b_n^2\|Le_n\|^2$*; coherent (smooth) residuals require the full matrix form. (Proof: Appendix F.)*

The hypotheses are not decorative: a gain estimate computed from the *same* noisy record carrying $\xi$ — the generic blind situation — generically violates (H1)–(H3), adding an $L\,m_\xi$ bias term and a $C_{\delta\xi}$ cross-term (Remark F.1.1). The identity is accordingly validated in the injected-residual (exogenous) regime (E4, Sec. 9) and applies to cross-fitted estimates restoring (H1)–(H2) by construction; for same-record blind pipelines it is exact only up to those two terms.

The Poisson plug-in $\Sigma_\xi = \mathrm{diag}(\tau_{G,n}^2 + b_n^2/\Phi_n)$ is exact at all photon counts *after the post-calibration substitution* $\hat{a}_n \to a_n$; before it both noise variances carry the factor $(a_n/\hat{a}_n)^2$ (Appendix F). With that qualifier, the identity — not an approximation — is what Fig. 9 tests. One metric convention matters: the identity is stated for the raw error $\mathbb{E}\|\hat{T}-T\|^2/S_2$, whereas the relMSE *measured* in Fig. 9 and in the reconstruction-bridge experiment (Fig. 9) uses least-squares scale alignment (Sec. 2), which can pull $\mathrm{relMSE}/v$ below 1 for DC-dominated objects (Sec. 9); the *raw* identity is verified directly by the injected coherent-residual experiment E4 (Supplementary Material S2), while Fig. 9 reads the scale-aligned image metric. The two metrics are *not* proportional (Appendix F.4, eq. (F.17)), so any prediction–observation comparison, in particular Prop 3's, must use the same metric on both sides. Specializing $L$:

- **Orthogonal / SRHT full inversion:** $B_L = 1$, so $\mathrm{relMSE} = v + (1/S_2)\sum_n(\tau_{G,n}^2 + b_n^2/\Phi_n)$: residual gain passes through at unit leverage.
- **Pairwise (differenced) Hadamard:** gain term $R_{\mathrm{pair}} = (K_{\mathrm{eff}}/4)\,D_H(T)\,\mathbb{E}[\Delta^2]$ — the second moment $\mathbb{E}\Delta^2$, not $\mathrm{Var}(\Delta)$, unless $\mathbb{E}\Delta = 0$ is separately assumed (a deterministic mismatch $\Delta \equiv \epsilon$ has zero variance but nonzero risk; Appendix F.3.2) — with $D_H \approx 1$ for non-DC coefficients and $\mathbb{E}[\Delta^2] \approx 2s^2\,r(\rho_{\mathrm{pair}})$ the adjacent-pair increment for OU drift ($r(\rho) = 1 - e^{-\rho}$; $r(\rho) \approx \rho$ for small $\rho$): differencing converts *slowness* directly into suppression.
- **Random / DGI:** $\mathrm{relMSE_{DGI}} = C_0^{\mathrm{floor}}/N + (1 + C_0^{\mathrm{lev}})v/N +$ noise terms. Three constants of the *same* pipeline must be kept apart (Appendix F.3.3): the fixed-design sampling floor $C_0^{\mathrm{floor}}$ (raw $\approx 10^3$ here, aligned $\approx 7\times10^2$ — a factor-1.54 metric difference), the drift leverage $\Lambda_{\mathrm{lev}} = NB_L = 1 + C_0^{\mathrm{lev}}$, and the population moment constant $K + \beta_4 - 2$. Mean subtraction removes the $K_{\mathrm{eff}}K(\mu/\sigma)^2$ background penalty from the floor but **not** from the leverage — drift modulates the large DC background coherently — so for mean-subtracted pipelines $C_0^{\mathrm{lev}}$ is the nonnegative-background value of (F.10), roughly $10^3\times$ the floor. There is **no universal** $C_0$: each constant depends on pattern statistics, background, normalization, *and metric* — for our SCGI-style testbed [7] the measured values are used throughout Fig. 9 and Fig. 4.

An honest corollary the identifiability results alone would obscure: **at equal residual variance** $v$**, orthogonal inversion is the best conduit** ($B_L = 1$, no $C_0/N$ estimator floor). The randomized bases win not by transmitting gain error more gracefully, but because Theorem B and the Sec. 6 window estimator let $v$ *itself* be driven down blindly, at rate $1/W$, where an ordered scan offers no blind handle on $v$ at all.

These closed forms compete, producing a finite-$N$ flip boundary.

**Prop 3 (finite-$N$ flip boundary; fixed-point form).** *All risks are the* raw *relative MSEs of Theorem 1.* $\rho^* = \inf\{\rho_{\mathrm{pair}} \ge 0 : R_{\mathrm{pair}}(\rho_{\mathrm{pair}}) \ge R_{\mathrm{rand}}(\rho_{\mathrm{pair}}, N)\}$ *satisfies, in the small-drift regime, the* fixed-point equation

$$1 - e^{-\rho^*} = \frac{2}{K_{\mathrm{eff}}\,D_H\,s^2}\big[C_0^{\mathrm{floor}}/N + (1 + C_0^{\mathrm{lev}})\,v_{\mathrm{blind}}(\rho^*, N)/N + R_{\mathrm{DGI,noise}} - R_{\mathrm{pair,noise}}\big],$$

*where* $C_0^{\mathrm{lev}}$ *is the pipeline's drift-leverage constant (the nonnegative-background value of (F.10) for mean-subtracted DGI pipelines — Appendix F.3.3) and* $v_{\mathrm{blind}}$ *is an externally measured function of* $(\rho, N)$*. Because the right side moves with* $\rho^*$*, this is* not *an explicit solution: existence needs the sides to cross in range, uniqueness a single-crossing condition; only for a constant bracket* $Q_0$ *does it collapse to* $\rho^* = -\log(1 - Q_0)$ *(no finite crossing when* $Q_0 \ge 1$*). Its leading order is the heuristic* $\rho^* \approx 2C_0^{\mathrm{floor}}/(NK_{\mathrm{eff}}s^2)$*; for mean-subtracted pipelines "negligible* $v_{\mathrm{blind}}$*" means* $(1 + C_0^{\mathrm{lev}})v_{\mathrm{blind}} \ll C_0^{\mathrm{floor}}$*, far stronger than* $v_{\mathrm{blind}} \ll C_0^{\mathrm{floor}}$*. (Proof: Appendix F.)*

*Remark:* $\rho_{\mathrm{pair}}$ (adjacent-pair decorrelation, this section) and $\rho_{\mathrm{bw}}$ (normalized bandwidth, Sec. 4) both grow with drift speed but are not equal; every Fig. 4 axis declares its convention (Appendix F.4).

The bridge is validated term by term across bases, drift speeds, and photon budgets in Supplementary Material S2 (Fig. 9) — numerical evidence, on simulation, that Theorem 1 is an accounting identity rather than a bound. Figure 4 reports the equal-frame competition on the $(\rho_{\mathrm{pair}}, \sigma_a)$ grid: the empirical winner map, the SRHT-paired-versus-ordered-Hadamard crossover (a comparison Prop 3 does not rank), and the no-free-parameter skeleton test of Prop 3 on a gain-known random arm (Sec. 9.4); for the blind random+DGI arm the corrected leverage constant predicts — and the data confirm — no flip anywhere on this grid. Three regimes emerge. For **slow drift** ($\rho_{\mathrm{pair}} < \rho^*$), pairwise Hadamard is fine: differencing suppresses what little the gain moves, and orthogonal conditioning is kept for free. For **intermediate drift**, randomized carriers win — the anchor buys a small $v_{\mathrm{blind}}$ no ordered scan can reach; the above-floor winner is the

SRHT-paired pipeline, which keeps orthogonal conditioning and avoids the $C_0(\text{pipeline})/N$ estimator floor that bare i.i.d.-random/DGI pays. For **fast drift** ($\rho_{\text{bw}} \to 1$), every *blind algebraic* strategy hits the noise floor together — a statement about blind separation, not the physics: an externally calibrated or metered gain restores ordinary inversion. The chart is a pre-acquisition instrument: locate the drift spectrum and photon budget on it, and the go/no-go decision — and the basis — follow before the first frame is taken.

The measured constants and conventions behind Fig. 4 are collected in Table I.

## IX. Numerical experiments

The theory of Secs. 4–8 makes quantitative predictions — a stationarity dichotomy, a $W^{-1/2}$ estimation law with object dependence confined to $K_{\text{eff}}$, an exact relMSE identity, a flip boundary, and a photon-limited Fisher scaling — and this section tests each. A closing subsection integrates five later verification experiments: the tall-design rank threshold (Fig. 6), whitening power (Supplementary Material S1), the coherent-residual validation of Theorem 1 (S2), the $C_0$ audit, and oracle-to-blind baselines (S3).

**Protocol.** All results are simulations; no hardware data enter (Sec. 1). Comparisons use matched budgets — equal frames and equal total flux per arm. The drift is an Ornstein–Uhlenbeck log-gain trace with per-frame decorrelation rate $\rho$, the knob setting $\rho_{\text{pair}}$ of Sec. 8; within each seed a single trace is shared by every arm (identical nuisance realization; traces differ across seeds, drift s.d. 0.030–0.104). The Hölder-$\beta$ model of Theorems B–C is the *analysis frame*, the OU trace the *simulation protocol*: an OU/AR(1) path does not literally satisfy a $\beta = 1$ Lipschitz bound, and the windowed estimator is applied to it without smoothness-based tuning. Blind correction uses one estimator for all arms — the sliding-window-mean AGC realization of the Sec. 6 windowed estimator — with no per-arm tuning and no denoising or post-processing anywhere. The object panel comprises ten shared objects from a fixed generator — five synthetic MNIST-style digits, the letters A and L, a periodic vertical stripe (width $\approx$ image/8), a ring (radii $\approx$ 0.22/0.34 of the image span), and one procedural natural patch — each normalized to $[0, 1]$ and spanning $K_{\text{eff}} = S_1^2/S_2 = 59.8$–$864.3$, each run at five drift seeds (fifteen for the reconstruction bridge), with errors computed against ground truth on the object support mask after the least-squares scale alignment of Sec. 8; every experiment shares base seed 20240708. Metrics: blind gain error is the scale-aligned relative error (gain traces normalized to unit mean, optimal scalar alignment); mask contrast is GT-mask CNR; reconstruction error is the scale-aligned relMSE of Sec. 8. Full per-figure enumerables and pattern conventions: Supplementary Material S5.

**Carrier stationarity (Fig. 2).** This experiment applies a *variance-stationarity diagnostic* to the noiseless carrier $\{B_n\}$ before any gain is applied (8192 frames per arm, $64 \times 64$ objects); the Brown–Forsythe test checks chunk-variance equality, a consequence of the stationarity ($\star$) requires rather than a direct test of its windowed log-carrier means. The Brown–Forsythe test at $p < 10^{-3}$ rejects 7 of 10 objects for ordered Hadamard, with $p$ reaching $10^{-20}$, against 0 of 10 for both i.i.d. random ensembles (uncorrected per-test $p$-values; the strongest rejections clear a Bonferroni-adjusted threshold by many orders of magnitude). The permuted arms — randomly permuted paired Hadamard ($P_{\text{row}}$) and SRHT ($P_{\text{row}}+D$, Appendix E.6) — each reject 2 of 10: a *fixed* reordering draw does not guarantee temporal homogeneity for every object, since the randomization guarantees of Secs. 5 and 7 are probabilistic over draws — this residue is the theory's own prediction, not an anomaly. The pixel-level randomization of the SRHT analysis ($P_{\text{col}}$, $D$) is probed by the factorized ablation of Sec. 9.6. The Parseval invariant closes the loop: every exhaustively paired arm has carrier $\text{CV}_B = 1/\sqrt{K_{\text{eff}}}$ to about $10^{-6}$ relative accuracy, and the random arms match $(\sigma_I/\mu_I)/\sqrt{K_{\text{eff}}}$ within $\pm 2\%$ — the ordered pathology lives entirely in the temporal variance structure, not in the marginal spread (descriptive counts over the ten-object panel).

**Blind gain identifiability (Fig. 3).** This is the core evidence for Theorem B, run over nine arms including the raw, non-paired Hadamard arms that feed bare signed Walsh coefficients to the estimator (bucket convention (i), diagnostic-only); per-arm errors and slopes with two-way CIs: Supplementary Material Table III. Raw arms are scored at the same 2048-frame budget; the single-pass 1024-frame diagnostics are indistinguishable (Table III), so the failure is chronology, not budget. In Fig. 3 the chronology failure of Theorem A is made visible: under the ratio AGC the raw ordered arm's blind gain error is 0.94–0.99 across all averaging windows (median best-window 0.899) and does not decay; shuffling the raw frames improves this to 0.794 — permutation helps even here — but the arm remains far above the random and paired arms at 0.007–0.017, because without pairing the near-zero-mean carrier leaves the AGC ill-posed. Under the log-domain estimator faithful to Theorem B (windowed means of $\log R$, masked to the $R > 0$ record per assumption (iv)) — which agrees with the ratio AGC to about 2% on all stationary-carrier arms with the same qualitative ordering — the raw ordered arm still cannot fall below 0.71 (non-decaying) while raw shuffled improves to 0.136, still at least $12\times$ above every randomized arm. Within the variance-obeying arms, the slow-drift variance-segment slopes of $\log(\text{err})$ against $\log W$ all sit inside the $W^{-1/2}$ band ($-0.39$ to $-0.61$), while naturally ordered paired Hadamard is shallow at $-0.24$: differencing cancels the drift within pairs but leaves the ordered envelope. Normalizing by the theory floor $\sqrt{\text{CV}_B^2/W}$ collapses the panel: the paired arms sit at 0.40–0.41 and the random arms at 0.89–0.90, each bundle within $\pm 5$–$7\%$ across all ten objects, and the best-window error correlates with $K_{\text{eff}}$ at $-0.86$ to $-0.90$ (descriptive) — consistent, within this tested panel, with the object entering through $K_{\text{eff}}$ alone, as Proposition 1 predicts to leading Gaussian/log-variance order. The shuffled-paired arm performing like the random arms bears out the permutation analysis of Sec. 7: pairing cancels the slow drift within pairs, and the permutation restores the stationarity that the ordering destroyed.

The 20-seed uncertainty replication (31,600 rows) confirms

TABLE I

HEADLINE MEASURED CONSTANTS BEHIND FIG. 4 (SECONDARY CONSTANTS AND GRID CONVENTIONS: SUPPLEMENTARY MATERIAL TABLE II). EVERY CONSTANT IS PIPELINE-SPECIFIC, AND FLOOR, LEVERAGE, AND MOMENT CONSTANTS OF THE *same* PIPELINE ARE DIFFERENT NUMBERS (APPENDIX F.3.3). CONFIDENCE INTERVALS: TWO-WAY (SEEDS-AND-OBJECTS) CROSSED CLUSTER BOOTSTRAP PERCENTILE INTERVALS OVER THE 20-SEED $\times$ 10-OBJECT REPLICATION, SLOPE FITS RESTRICTED TO $W \leq 16$; AN OBJECT$\times$SEED BOOTSTRAP TREATING THE 200 CELLS AS INDEPENDENT UNDERSTATES UNCERTAINTY; NO QUALITATIVE CONCLUSION CHANGES.

| Quantity | Value in this pipeline | Uncertainty |
|---|---|---|
| Drift leverage $\Lambda_{\text{lev}} = N\,B_L = 1 + C_0^{\text{lev}}$, via the bridge experiment's DGI leverage law. **Not** the Prop 3 floor $C_0^{\text{floor}}$ (raw $\approx 1.0$–$1.1 \times 10^3$, aligned $\approx 6.2$–$7.2 \times 10^2$; Sec. 9.4) | $\Lambda_{\text{lev}} \in [7.38 \times 10^5,\ 7.68 \times 10^5]$ across the $N$ sweep (4% drift) | Monte Carlo mean over the bridge experiment's residual probes (10 objects $\times$ 15 seeds per $N$), not deterministic; no bootstrap CI |
| $D_H(T)$ (pairwise-Hadamard factor) | measured over the ten-object panel: min/median/max $= 0.971/0.988/0.999$ (per object via the analytic identity $D_H(T) = \frac{1}{K}\sum_k (1 - x_k^2)^2$, $x_k = c_k/S_1$) | deterministic per object (identity evaluated on the panel); no resampling needed |
| $v_{\text{blind}}$ (residual gain second moment after blind correction) | best-window blind gain error at $\rho_{\text{pair}} = 10^{-3}$, $s = 0.1$ (2048 frames, ratio AGC): srht_paired $0.0077$ $[0.0066, 0.0083]$; hadamard_random_paired $0.0073$ $[0.0063, 0.0079]$; random_binary $0.0150$ $[0.0130, 0.0165]$; log-domain estimator agrees to $\sim 2\%$ except randomly permuted paired Hadamard ($0.0119$ $[0.0090, 0.0133]$); $v_{\text{blind}}$ is *estimated by* the squared best-window scale-aligned gain error (a convention, not an identity; cf. (F.17)): $(\text{gain error})^2 \approx 5\times10^{-5}$–$2 \times 10^{-4}$ | 95% two-way clustered CIs shown inline (see caption) |
| Prop 3 boundary, no-free-parameter test (gain-known random arm), *raw metric on both sides* | 40/40 raw crossings ($\sigma_a \geq 0.1$), median factor **1.044**, max 1.200; raw floor $C_0^{\text{floor,raw,fix}} = 980.1$–$1114.5$; the former aligned factor 1.54 equals the raw/aligned floor ratio 1.5399 (metric mismatch, not a cubic remainder); $\sigma$-exponent $-2.11$ vs. $-2.07$ predicted; pair law (F.8) median 0.997 (evidence: `results/prop3_raw_metric_r1`, `results/prop3_nofreeparam_r1`) | descriptive agreement statistics; no bootstrap CI computed |

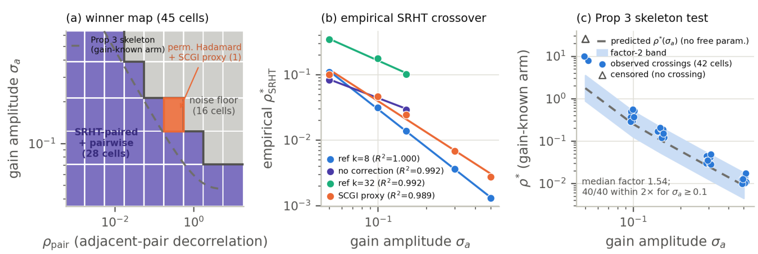


Fig. 4. Empirical winner map, SRHT crossover, and the no-free-parameter Prop 3 skeleton test at equal frame budget ($N = 2048$, $K = 1024$, OU drift, blind corrections only). (a) Winner per $(\rho_{\text{pair}}, \sigma_a)$ cell by mean PSNR among candidates passing the above-floor gate relMSE $< 0.5$ (grey: noise-floor cells; outlined: the single non-SRHT winner); dashed: the Prop 3 gain-known skeleton at median measured constants. (b) Empirical SRHT-paired-versus-ordered-Hadamard crossover under several correction references, with power-law fits — descriptive fits of a comparison Prop 3 does not rank. (c) Skeleton test (gain-known arm, $v = 0$) as originally run in the *scale-aligned* metric: predicted boundary with factor-2 band against the 42 aligned crossings; triangles: censored cells. The visible offset (median factor 1.54) is the raw/aligned metric mismatch of Sec. 9.4; the raw-vs-raw comparison gives 40/40 crossings at median factor 1.044 (max 1.200). No-flip outcome: Sec. 9.4 and Supplementary Material S4.

these slopes under the fixed $W \leq 16$ variance-segment convention: the randomized arms' median two-way slopes lie between $-0.39$ and $-0.62$, the naturally ordered paired arm's ratio-AGC slope CI straddles zero (no detectable decay), and the raw ordered arm stays essentially flat at $0.899$ (per-arm values and CIs: Table III). The two-way crossed cluster bootstrap widens naive object$\times$seed intervals by 3–5$\times$ without crossing any qualitative boundary (rationale: Table I caption). One nuance: under the log-domain estimator the naturally ordered paired arm shows a shallow decay ($-0.222$ $[-0.306, -0.054]$) where the ratio AGC shows none; both remain far outside the $W^{-1/2}$ band, so the chronology conclusion stands ("no detectable decay" attaches to the ratio AGC). Best-window blind gain errors at this operating point are in Table I; the exploratory slopes and $R^2$ above used an adaptive variance segment, and the two conventions agree on every qualitative conclusion.

**The reconstruction bridge (Supplementary Material S2, Fig. 9).** This experiment tests Theorem 1 as an identity rather than a bound, on a fully balanced 8100-row design (three reconstructors, $N \in \{1024, 2048, 4096\}$, six injected $v$, ten objects, fifteen seeds; predicted-versus-measured panel: Fig. 9). The deterministic inverses are flat in $N$, as $B_L = 1$ demands (per-$v$ log–log slopes at most $0.074$ orthogonal, $0.014$ SRHT — no $1/N$ reconstruction floor), while the random/DGI arm decays as $1/N$ at every $v$ (slopes $-1.031$ to $-0.974$) with a quantitative leverage law: measured relMSE agrees with leverage $\times\ v$ within $[0.91, 1.06]$, and leverage $\times\ N$ sits between $7.38 \times 10^5$ and $7.68 \times 10^5$ across the $N$ sweep (4% drift). For the orthogonal inverse, $\text{relMSE}/v \approx 1.0$ for 8 of 10 objects; the two DC-dominated objects (natural_patch $0.35$, stripe $0.54$) fall below the $v$ line because the scale alignment absorbs the gain component co-linear with the object — an alignment effect, not a failure of the identity. The metric is scale-aligned, with the deterministic and random/DGI arms judged against different conventions (unit leverage versus the leverage $\times N$ law), so cross-arm error levels are not directly comparable; the raw-MSE identity-verification role is carried by the E4 comparison below.

**Phase diagram and ablation (Fig. 4; ablation panels in Supplementary Material S1, Fig. 8).** The phase diagram sweeps the drift rate $\rho \in [10^{-3}, 10]$ and noise level $\sigma \in [0.05, 0.5]$ at equal frames, with an above-floor gate $\mathrm{relMSE} < 0.5$ — a conservative meaningfulness cut excluding reconstructions whose error energy is at least half the object energy; cells failing it are reported as noise floor rather than assigned a winner. The winner map has 45 cells (9 drift rates $\times$ 5 noise levels), of which 29 are above floor: SRHT-paired acquisition with pairwise correction wins 28 of the 29; one cell ($\rho_{\mathrm{pair}} = 0.3$, $\sigma_a = 0.15$) goes to randomly ordered pairwise Hadamard with the SCGI-proxy correction. The boundary curves shown are two-parameter power-law fits ($R^2 = 0.989$–$0.9995$, fitted exponents $-0.94$ to $-1.93$) to the empirical SRHT-paired-versus-ordered-Hadamard crossovers — a descriptive summary of a comparison that Prop 3 explicitly does not rank, not a test of Prop 3's closed form. The no-free-parameter test of Prop 3 is reported separately, *with the raw metric on both sides*: on the same grid, with every constant measured and a gain-known random arm ($v = 0$), the raw-floor prediction $\rho^* = -\log\left(1 - 2(C_0^{\mathrm{floor,raw,fix}}/N)/(K_{\mathrm{eff}} D_H s^2)\right)$ lands within a **median factor 1.044 (max 1.200) of the 40/40 observed raw crossings** at $\sigma_a \geq 0.1$ (evidence: `results/prop3_raw_metric_r1`, same OU paths as the published grid, no new simulation). The previously reported aligned-metric factor 1.54 is a *raw/aligned metric mismatch* (the raw/aligned floor ratio is 1.5399), not a small-mismatch remainder effect (Table I). For the blind random arm, the residual-gain leverage in (F.12) must carry the nonnegative-background $\Lambda_{\mathrm{lev}} = 1 + C_0^{\mathrm{lev,raw\ background}}$ of (F.10) ($\approx 10^6$), not a floor constant ($\approx 10^3$) — the mean-subtraction asymmetry of Sec. 8, a $\sim 1.4 \times 10^3$ correction verified to 5% at white drift. With the corrected constant the theory predicts — and the data confirm, 100/100 cells — that blind random+DGI never overtakes ordered pairwise Hadamard anywhere on this grid (per-cell margins: Supplementary Material S4); the intermediate-drift winner is SRHT-paired acquisition, which avoids the $C_0/N$ floor entirely. The ablation (Supplementary Material S1, Fig. 8) isolates the mechanism at the single above-floor drift point $\rho = 10^{-3}$: SRHT gains +5.45 dB over ordered Hadamard, pixel-sign randomization alone +5.47 dB, acquisition-order permutation alone +5.39 dB — each ingredient by itself recovers essentially the full advantage, and combining them is not additive (descriptive values at this grid point). Its arms randomize acquisition time only ($P_{\mathrm{row}}$, $P_{\mathrm{row}}+D$; the pixel-permutation attribution is Sec. 9.6): what matters empirically is that the chronology is randomized, not the particular form of the randomization. At the largest simulated decorrelation rates ($\rho \geq 1$) all blind variants approach the noise floor within $\pm 0.09$ dB — consistent with the information-loss intuition behind the $p \to N$ limit of Theorem A$'$, though not a direct test of the low-pass threshold unless $\rho_{\mathrm{bw}}$ and $\rho_{\mathrm{pair}}$ are controlled separately.

**Low-photon robustness (Fig. 5).** This experiment tests the soft-log analysis of Sec. 6 under Poisson counting, with *five* estimator arms on the same paired Poisson draw at each budget: naive clipped log, Anscombe, the uncalibrated soft-log proxy, the mean-Poisson calibrated proxy $m_\alpha^{-1}(\mathrm{mean}_W\, \psi_\alpha)$, and the oracle-known-carrier calibrated soft-log of Theorem C (per-window carrier-calibration root, carriers $b_k$ from the simulation truth — a flat-field benchmark; requested $W = 64$, effective interior width 65; truncated edge windows; Appendix D.4); each point aggregates 50 object-by-seed cells, error bars the SD over those cells. Every ratio below is a log-domain (centered-log-gain) mean-MSE ratio, numerator over denominator as named — the metric comparable to the oracle common-shift diagnostic $\mathrm{mean}_n[1/I_n]$ of Appendix D.4 ($d = 0$). At mean counts $\bar{\lambda} \leq 1$ the naive clipped log is 24.9–28.7$\times$ worse than the uncalibrated proxy, 16.3–21.0$\times$ worse than Anscombe, and 1.8–11.4$\times$ worse than the calibrated estimator (all naive-over-method ratios). The two calibrated arms track the $1/(W\bar{\lambda})$ law across the variance regime — slopes $-1.000$ (oracle-carrier) and $-0.99$ (mean-Poisson proxy, overlapping the oracle arm to 0.4% at this protocol's small carrier variation) over $\bar{\lambda} \in [2, 32]$, against $-0.920$/$-0.732$ for the uncalibrated proxy (its $[1, 16]$ fit is contaminated by its own bias transition). At budgets $\bar{\lambda} \leq 2$ the uncalibrated proxy and Anscombe fall *below* the common-shift diagnostic — shrinkage bias, not super-efficiency (MSE above it likewise does not certify unbiasedness; Appendix D.4) — whereas the calibrated estimator stays above it at every tested budget, within 2–18% at $\bar{\lambda} \leq 1$ and 14–31% through $[2, 32]$. The proxy's apparent 1.19–1.53$\times$ advantage over Anscombe at these budgets is that shrinkage bias, not efficiency; and the calibrated estimator loses to Anscombe in mean MSE at $\bar{\lambda} \leq 1$ (Anscombe 1.8–8.9$\times$ better) — a bias-bought advantage (Sec. 6 trade-off) — overtaking it only for $\bar{\lambda} \geq 32$, where the pairwise ratios converge to 1.00 ($\pm 4\%$). At the high-photon end the floor is drift-limited, not photon-limited: $2.434 \times 10^{-4}$ ($\rho = 10^{-3}$) $\to$ $1.411 \times 10^{-4}$ ($\rho = 10^{-4}$) for the proxy arms, $2.168 \times 10^{-4} \to 1.125 \times 10^{-4}$ for the oracle-carrier estimator — 10.92% lower at $\rho = 10^{-3}$ and 20.28% at $\rho = 10^{-4}$ (two distinct margins), since it deconvolves the known carrier (floor probe: Supplementary Material S4, Fig. 12).

**Threshold, whitening-power, and accountability verifications (Fig. 6 and Supplementary Material S1–S4).** Five further experiments close the remaining theory–evidence gaps; unless a bootstrap interval is stated, every number below is a descriptive result of its grid.

*Tall-design threshold scan (Fig. 6).* A direct test of the local component of Theorem A$'$ at two object sizes ($K = 64$: 20 seeds per cell, 1300 cells; $K = 128$: 30 seeds per cell, 1950 cells), over five low-pass drift dimensions $p \in \{3, 5, 9, 17, 33\}$ at margins $N - K - p \in [-8, +16]$. At both sizes the local rank test on the collision Jacobian transitions from 0% to 100% pass *exactly* at margin $N - K - p = -1$ for every $p$ (3250 cells, no exceptions) — matching the proved a.e. rank formula $\operatorname{rank} D\Phi_M = \min\{N, K + p - 1\}$ of Theorem A$'$(i); below the wall the observed rank deficits correspond, by the same theorem, to genuine positive-dimensional alias families, not numerical degeneracy. Being a rank statement, the scan

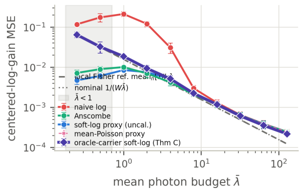


Fig. 5. Low-photon robustness under Poisson counting, on identical random-basis carriers and shared gain traces (soft-log $\alpha = 0.5$, requested $W = 64$ / effective interior width 65, $\rho_{\text{pair}} = 10^{-3}$, $s = 0.1$; truncated edge windows). Five methods — naive clipped log, Anscombe, uncalibrated soft-log proxy, mean-Poisson calibrated proxy, oracle-known-carrier calibrated soft-log (Theorem C) — on the same paired Poisson draws; error bars: SD over the 50 object-by-seed cells. Centered-log-gain MSE ($d = 0$) versus $\bar\lambda$ against the *oracle common-shift diagnostic* $\text{mean}_n[1/I_n]$ (dashed; dotted: nominal $1/(W\bar\lambda)$; shaded: $\bar\lambda < 1$) — not a nuisance-path quotient CRB. The calibrated arms (diamonds: oracle-carrier root; the mean-Poisson proxy overlaps it) track the $1/(W\bar\lambda)$ law and stay above the diagnostic; the uncalibrated proxy and Anscombe buy low-count MSE with shrinkage bias (below the diagnostic at $\bar\lambda \leq 2$). Ratios, slopes, floor: Sec. 9.5.

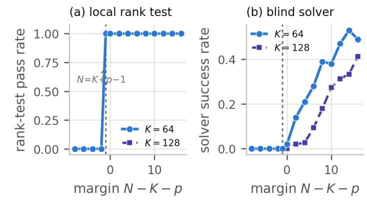


Fig. 6. Tall-design threshold scan (local-rank test of Theorem A$'$(i) plus an algorithmic-recovery probe) at $K = 64$ (20 seeds/cell, 1300 cells) and $K = 128$ (30 seeds/cell, 1950 cells), $p \in \{3, 5, 9, 17, 33\}$, margins $N - K - p \in [-8, +16]$, all $p$ pooled. (a) The rank test transitions $0\% \to 100\%$ *exactly* at margin $-1$ at both sizes (3250 cells, no exceptions), verifying $N = K + p - 1$. (b) The blind alternating solver starts at the same wall but climbs slowly — algorithmic recovery, not uniqueness; rates in Sec. 9.6.

verifies the local component only, not the global sufficiency thresholds of A$'$(ii)–(iii) or their conjectured sharpness. The companion solver panel (blind alternating recovery, 8 restarts $\times$ 150 iterations) starts at the same wall but climbs slowly — about 2% success at margin 0 and $\sim 50\%$ at margin $+16$ for $K = 64$ ($\sim 41\%$ for $K = 128$) — an *algorithmic-recovery* observation, not a uniqueness test: uniqueness and recovery separate, as Sec. 2's third notion anticipates.

*Multi-permutation whitening power and factorized randomization ablation (Supplementary Material S1, Fig. 7).* Two randomization semantics must be kept separate (Appendix E.6): the pixel-level randomization of the SRHT theory ($P_{\text{col}}$, $D$; acquisition-order-invariant by Lemma E.1) and the acquisition-order reordering $P_{\text{row}}$, which is what the Brown–Forsythe screen probes. The factorized ablation attributes the whitening-power headline: either permutation alone substantially reduces the rejection rate — from 70% (ordered) to 5.6% ($P_{\text{col}}$) and 12.2% ($P_{\text{row}}$), the theory's $P_{\text{col}}{+}D$ to 11.9% in natural order — a large but incomplete suppression (all three stay well above the $\sim 0.1\%$ nominal rate of the $p < 10^{-3}$ test), so Appendix E's whitening claims do not depend on the row-order confound; pixel signs alone barely move it (43.4%). The per-draw Walsh certificate fails (Spearman 0.092, $p = 0.10$): an honest null, consistent with the probabilistic nature of permutation whitening (Sec. 7).

*Coherent-residual validation of Theorem 1 (E4, Supplementary Material S2, Fig. 10).* Under five injected residual-covariance structures (150 cells, raw identity-verification metric), the matrix law of Theorem 1 tracks the measured relMSE within 4.2%, while the scalar $v\,B_L$ law — exact for the orthogonal and SRHT inversions — fails by up to 3.65$\times$ for random/DGI under low-pass-coherent residuals: a direct verification of the coherent clause. E4 uses *injected* residuals; it does not probe the same-record $\delta$–$\xi$ dependence flagged after Theorem 1.

*$C_0$ audit and oracle-to-blind baselines (Supplementary Material S3, Fig. 11).* The measured one-sample constant agrees with the Appendix-F moment formula to 0.35% mean (2.01% worst) relative error over four ensembles $\times$ five objects, the ensemble spread confirming that $C_0$ is pipeline-specific; and on four held-out objects the ordering static $\geq$ oracle $\geq$ blind-AGC $\geq$ no-correction holds in CNR for 4 of 4, with an oracle$-$blind gap of only 0.035–0.147: the blind estimator concedes little to a gain oracle; forgoing correction costs the most.

## X. Discussion: significance, scope, and outlook

The results of Sections 2–9 are pre-acquisition design tools, evaluated on the illumination basis and the expected drift *before* photons are collected. Four deliverables follow, then limits and the invited experimental program.

**A go/no-go identifiability predictor.** The tall-design thresholds of Theorem A$'$ convert "will this acquisition self-calibrate?" into arithmetic — local identifiability iff $N \geq K + p - 1$, generic exact at $N \geq K + p$, uniform at $N \geq 2K + p - 1$, for any fixed drift class $S \ni \mathbf{1}$ including the physical low-pass family (proved sufficiency directions; sharpness of the two global thresholds remains a labeled prediction). Through $p \approx \rho_{\text{bw}} N$, slow drift needs only modest oversampling, while $\rho_{\text{bw}} \to 1$ forces the statistical anchor of Theorem B (Sec. 4) — which itself degrades once the gain decorrelates within the averaging window (the $\rho \geq 1$ convergence in Fig. 8).

**An SRHT design rule.** When blind slow gain is the operative nuisance, do not run ordered Hadamard: its chronology hands the object a temporal signature indistinguishable from drift. Randomize the carrier per the Sec. 7 rule, keeping the two permutation semantics distinct (Appendix E.6): $P_{\text{col}}$ enters the covariance-whitening guarantee, $P_{\text{row}}$ the exchangeability anchor of Sec. 5; empirically either permutation alone substantially reduces the non-stationarity screen's rejections, signs alone barely (Sec. 9.6). Retain a positivity/offset margin and choose $W$ from the bias–variance tradeoff: statistical self-calibration without surrendering orthogonal conditioning.

**A design map that was exposed to falsification.** Theorem 1 and the flip boundary $\rho^*$ (Prop 3) render relMSE across DGI, SRHT, and pairwise-Hadamard as a chart over basis, drift, and noise (Fig. 4); $\rho_{\text{pair}}$ and $\rho_{\text{bw}}$ are distinct axes and must be labeled as such before any overlay (Sec. 8). The boundary prediction met a two-regime outcome (Sec. 9.4): the gain-known skeleton validated with zero fitted parameters at median factor 1.044 once prediction and observation share the raw metric (the earlier factor 1.54 was a raw/aligned metric mismatch, not physics), and the blind-arm flip prediction as originally written falsified and repaired through one measurable constant

— the drift leverage must carry the nonnegative-background $\Lambda_{\mathrm{lev}}$, after which theory predicts and data confirm no flip for the mean-subtracted blind pipeline. Both failures were bookkeeping (a constant, a metric), not the functional form — sharpening the rule that every constant is pipeline-specific *and* metric-specific, with floor and leverage constants of the *same* pipeline three orders of magnitude apart.

**A cross-domain principle.** The exportable statement is basis-agnostic: *to self-calibrate a multiplicative nuisance, design the carrier so its distribution is stationary in the nuisance coordinate.* Blind gain/phase calibration, flat-field correction, coded-aperture and computational microscopy, and single-pixel spectroscopy share this bilinear structure — analogues and motivation, not proven domains; each needs its own operator analysis of the diagonal-row map [14], [13].

**Scope and limits.** The claims are theory and simulation only, with no hardware validation. Uniqueness near $N = K+p$ is an identifiability statement, not a conditioning guarantee (stability is a separate random-matrix question), and sharpness of the two global thresholds remains a labeled prediction (Sec. 4). Random square unconstrained designs are *not* algebraically identifiable (Theorem A). The soft-log avoids the log-zero singularity but cannot manufacture information: zero photons carry zero Fisher information — an information limit, not an estimator defect.

**Experimental feasibility.** The theory issues sharp, falsifiable bench predictions under programmable drift: the gain-known flip point $\rho^*$, the no-flip prediction for mean-subtracted blind DGI, the SRHT-versus-ordered crossover, and the Fig. 3 blind gain-error separation; hardware tests are left to future work with an optical-imaging group.

In conclusion: the square-design obstruction and tall-design thresholds (Theorems A, A′), the stationarity anchor with its minimax-rate-optimal estimator (Theorems B–D), the SRHT synthesis (Sec. 7), and the finite-noise bridge with its flip boundary (Theorem 1, Prop 3) make gain observable not by measuring it directly, but by making the object stop pretending to be time — an avenue for self-calibrating acquisition in dynamic scattering media.

## Supplementary Material

The supplementary document contains: S1 — permutation-whitening power and the factorized randomization ablation; S2 — the reconstruction-bridge panel (Fig. 9) and the coherent-residual (E4) validation of Theorem 1; S3 — the $C_0$ audit and oracle/metered/blind baselines; S4 — extended flip-boundary detail, the drift-floor probe, and replication tables; S5 — the shared simulation protocol; Appendices A–F — the complete proofs. Pointers "Appendix X" refer to the supplementary document; its reference list shares this one's numbering.

## Code and Data Availability

All figures and tables in this paper and its supplement are reproducible from the project repository (experiment runners, committed result CSVs, `make_pub_figures.py`); complete code and result data are publicly available at https://github.com/ccyyyYyzz/GI_a1 (branch `scgi-ceiling-diagnostic-r1`).

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